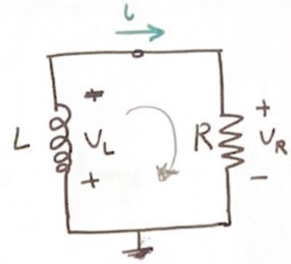
The Source-Free RL Circuits.

At $t=0$, assume the inductor has an initial current I_0;



Apply KVL

$$V_L + V_R = 0 \Rightarrow L\frac{di}{dt} + Ri = 0$$

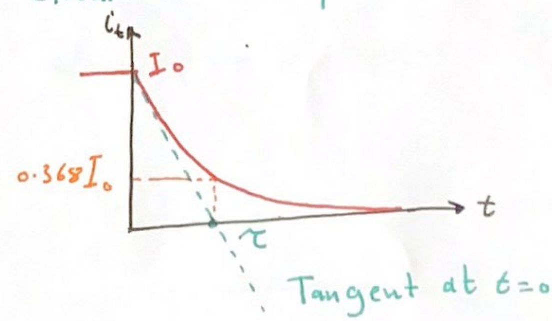
$$\frac{di}{dt} = -\frac{R}{L}i$$ (arranging variables for solving the differential equation using separation method)

$$\int_{I_0}^{i_t} \frac{di}{i} = -\int_0^t \frac{R}{L}\,dt$$ [Integrate both side, $I_0 \to i_{(t)}$, $0 \to t$]

$$\ln i_t - \ln I_0 = \frac{-R}{L}t \Rightarrow \ln\frac{i}{I_0} = \frac{-R}{L}t$$ [Power of e]

$$\frac{i}{I_0} = e^{\frac{-R}{L}t} \Rightarrow \boxed{i = I_0\, e^{\frac{-t}{L/R}} \qquad \tau = \frac{L}{R}}$$

The natural Response of the RL circuit is an exponential decay of the initial current.



For the Resistor, to find V_R, P, W_R

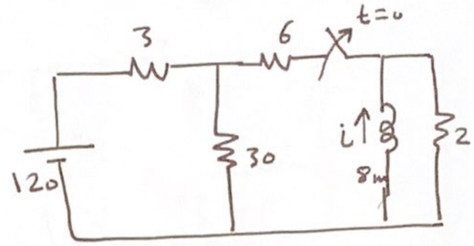
$$V_R = iR = I_0 R\, e^{-t/\tau}$$

$$P = V_R i = I_0^2 R\, e^{-2t/\tau}$$

$$w_R = \int_0^t P_\tau\, d\tau = \frac{1}{2} L I_0^2 (1 - e^{2t/\tau})$$

As $t \to \infty$, $W_R(\infty) = \frac{1}{2} L I_0^2 = W_L(0)$, the energy initially stored in the inductor is eventually dissipated in the resistor.

Ex: The switch has been closed for a long time and is opend at $t=0$. Find



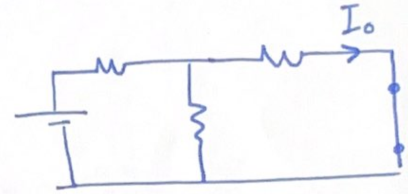
a) I_0

b) $W_L(0)$

c) $i_{(t)}$

d) Inductor energy dissipated in 2Ω after 5 ms. percentage to the inital.

For $t<0$, the inductor act as short circuit



1- find I_{Total}

$$R_{6\|30} = \frac{\overset{5}{\cancel{30}} \cdot \cancel{6}}{\underset{\cancel{6}}{\cancel{36}}} = 5\ \Omega$$

$$I_T = \frac{120}{8} = 15\ A$$

$$I_6 = I_L = \frac{30 \cdot 15}{30+6} = 12.5\ A \quad \text{or} \quad I_0 = -12.5 A \text{ (opposit direction)}$$

b) $W_L(0) = \frac{1}{2} L I_0^2 = \frac{1}{2} \cdot 8 \cdot 10^{-3} \cdot (-12.5)^2 = 0.625\ J$

c) $i_{(t)} = I_0\, e^{-t/\tau}$, $\tau = \frac{L}{R} = \frac{\overset{4}{\cancel{8}} \cdot 10^{-3}}{\cancel{2}} = 0.004$, $\frac{1}{\tau} = 250$

$$i_{(t)} = -12.5\, e^{-250t}$$

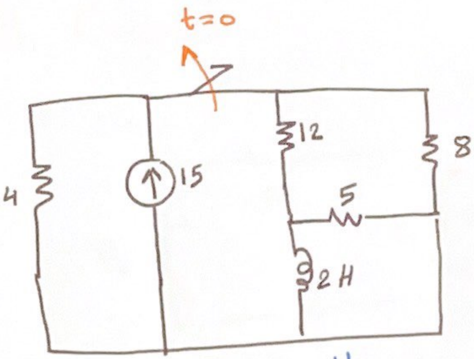
d) $i_5 = -12.5\, e^{-250(0.005)} = -3.58\ A$

$$W_5 = \frac{1}{2} L\, (-3.58)^2 = 51.3\ mj$$

$W_{dis} = 625 - 51.3 = 573$

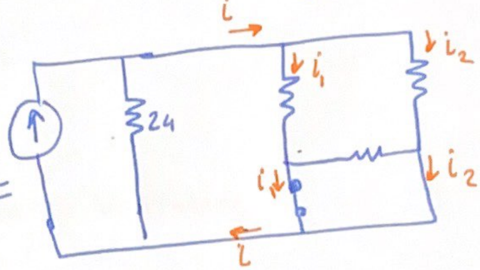
$\% = \frac{573}{625} \cdot 100 = 91.8\%$

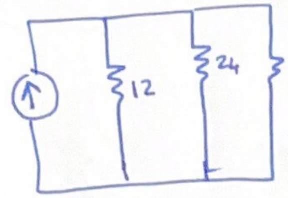
Ex: Find $i_{(t)}$ for $t \geq 0$



1- Find the initial current for the inductor

$i_{12} = i_{0_L}$, R_5 is short.





$$R_{24||8} = \frac{\cancel{24}\cdot 8}{\cancel{32}_{4}} = 6$$

$$I_{12} = \frac{15\cdot 6}{12+6} = \frac{15\cdot\cancel{6}}{\cancel{18}_{3}} = 5\text{ A}$$ initial current

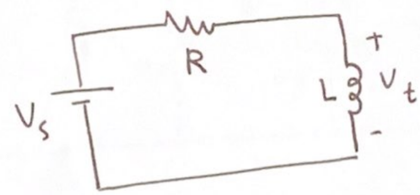
[2] For $t \geq 0$ $R_T = R_{5}||(12+8) = \frac{\cancel{5}\cdot 20}{\cancel{25}_{5}} = 4$

$$\tau = \frac{L}{R} = \frac{2}{4} = \frac{1}{2}$$

$$i_{(t)} = 5\,e^{-t/0.5} = 5e^{-2t}\text{ A}$$

The Step Response of an RL Circuit

Applying KVL



$$V_s = Ri + L\frac{di}{dt}$$

$$V_R + V_L = V_s$$

$$\frac{di}{dt} = \frac{V_s - Ri}{L} = \frac{V_s}{L} - \frac{R}{L}i$$

$$\frac{di}{dt} = \frac{-R}{L}\left(i - \frac{V_s}{R}\right)$$ separate Variables

$$\int_{I_0}^{i} \frac{di}{\left(i - \frac{V_s}{R}\right)} = \int_0^t \frac{-R}{L}\,dt$$ Integrate both side $I_0 \rightarrow i_t$, $0 \rightarrow t$

$$\ln\left(i - \frac{V_s}{R}\right)\Big]_{I_0}^{i} = \frac{-R}{L}t\Big]_0^t \implies \ln\left(i - \frac{V_s}{R}\right) - \ln\left(I_0 - \frac{V_s}{R}\right) = \frac{-R}{L}t$$

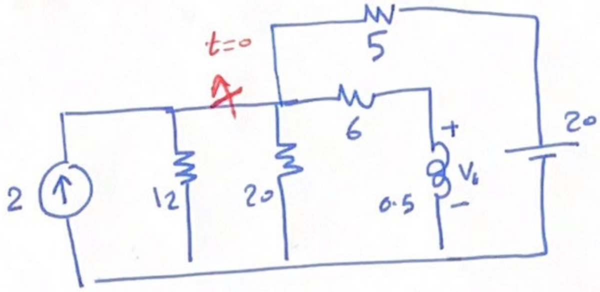
$$\ln\frac{\left(i - \frac{V_s}{R}\right)}{\left(I_0 - V_s/R\right)} = \frac{-R}{L}t \Rightarrow \frac{i - V_s/R}{I_0 - V_s/R} = e^{-R/L\, t}$$

$$\boxed{i = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right)e^{-(R/L)t}}$$

$\frac{V_s}{R} = i_{s.s}$, $\left(I_0 - \frac{V_s}{R}\right)e^{(-R/L)t}$ = transient current

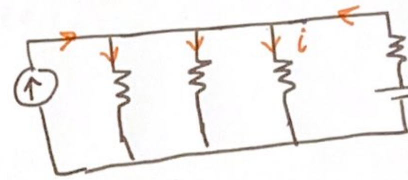
$$i = i(\infty) + \left(I_0 - i(\infty)\right)e^{-t/\tau} \qquad \tau = \frac{L}{R}$$

7.56 Find $V_{(t)}$ for $t > 0$



For $t < 0$ the switch is closed, and the inductor act as short circuit.

1- Find i through $R_6 = i_L$



Using KVL

$$\frac{V}{12} + \frac{V}{20} + \frac{V}{6} + \frac{V-20}{5} = 2$$

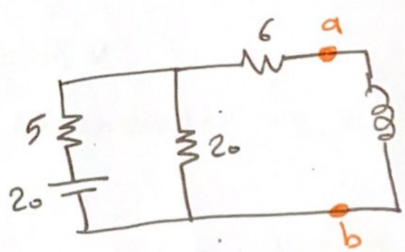
$$V\left(\frac{1}{12} + \frac{1}{20} + \frac{1}{6} + \frac{1}{5}\right) = +2 + \frac{20}{5}$$

$$V\,(0.0834 + 0.05 + 0.166 + 0.2) = \frac{10+20}{5}$$

$$0.5\,V = 6 \quad \Rightarrow \quad V = 12\ V$$

$$I_o = \frac{V}{R_6} = \frac{12}{6} = 2\ A$$

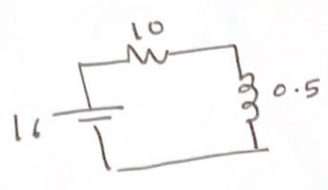
2- After opening the switch
Find R_{Th}, V_{Th} across a, b



R_{Th} $\quad (5 \| 20) + 6$

$$\frac{5 \cdot 20}{25} + 6 = 10\ \Omega$$

$$V_{Th} = V_{R_{20}} = \frac{20 * 20}{25} = 16\ V \quad \text{(V. divider)}$$



$$i = \frac{V_s}{R} + \left(I_o - \frac{V_s}{R}\right) e^{-t/\tau}$$

$$\tau = \frac{L}{R} = \frac{0.5}{10} = \frac{1}{20} = 0.05 \quad , \quad \frac{V_s}{R} = \frac{16}{10} = 1.6\ A$$

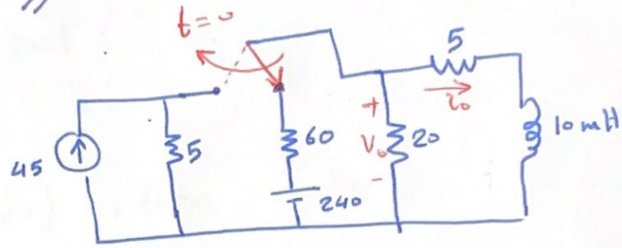
$$i = 1.6 + (2 - 1.6)\, e^{-t/0.05}$$

$$i = 1.6 + 0.4\, e^{-20t}$$

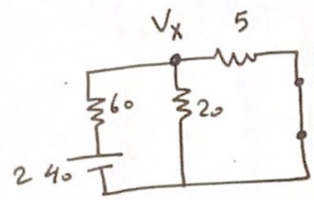
$$V_t = L \frac{di}{dt} = 0.5 \left[0.4\, e^{-20t} \cdot -20 \right]$$

$$= -4\, e^{-20t}$$

Ex: Find $i_o(t)$ and $V_o(t)$ at ≥ 0



At $t < 0$

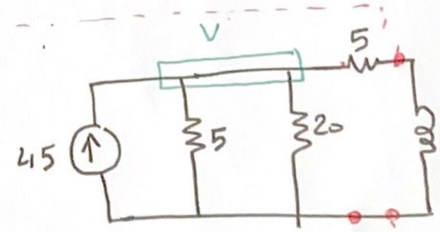


$$\frac{V_x - 240}{60} + \frac{V_x}{20} + \frac{V_x}{5} = 0 \Rightarrow \frac{V_x - 240 + 3V_x + 12V_x}{60} = 0$$

$16 V_x = 240$ and $V_x = \frac{240}{16} = 15\ V$

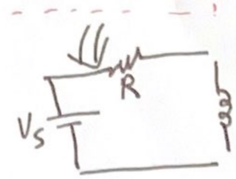
$i_o = \frac{V_x}{5} = \frac{15}{5} = 3\ A$ initial current on the inducter at $t \geq 0$

at $t > 0$



1- Find R_{Th} on L

$$R_T = (5 \| 20) + 5 = \frac{\cancel{5} \cdot \cancel{20}^{4}}{\cancel{25}_{\cancel{5}}} + 5 = 9\,\Omega$$

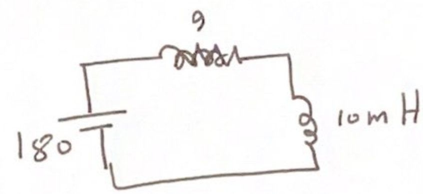


2- V is the voltage across L

$$\frac{V}{5} + \frac{V}{20} = 45 \Rightarrow \frac{4V + V}{20} = 45$$

$\frac{\cancel{5}V}{\cancel{20}_{4}} = 45$ and $V = 180\ V$

$$i_{(t)} = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right) e^{-t/\tau}$$



RL series circuit

$$\tau = \frac{L}{R} = \frac{10\,m}{9} = 1.112\ mS.$$

$$\frac{1}{\tau} = 900$$

$$i_{(t)} = \frac{\cancel{180}^{2}}{\cancel{9}} + \left(3 - \frac{180}{9}\right) e^{-900t}\ A$$

$$i_t = 20 - 17e^{-900t}\ A$$

$V_0 =$ Voltage of (5Ω + inductor) as they parallel

$$= 5i_0 + L\frac{di}{dt}$$

$$= 5\left(20 - 17e^{-900t}\right) + 10 \times 10^{-3}\left(-17e^{-900t} \cdot -900\right)$$

$$= 100 - 85e^{-900t} + 153e^{-900t}$$

$$V_0 = 100 + 68e^{-900t}\ V$$